CMS MEDICARE INPATIENT EXECUTIVE SUMMARY

Audit Risk Analysis — Medicare Inpatient Billing Anomalies

Data Year: 2023 | Published: July 2025

Executive Brief

This report presents findings from an analysis of Medicare Inpatient Prospective Payment System (IPPS) billing data for fiscal year 2023, covering **146,251 provider-DRG records** across **2,937 distinct hospital providers** nationwide. The analysis identifies systematic billing anomalies through charge-to-payment ratio analysis, statistical outlier detection, and a composite audit risk scoring model. Findings support prioritised allocation of compliance review resources ahead of the next CMS reporting cycle.

146,251	2,937	5.52x	141.52x	5.36%
Provider-DRG Records	Distinct Providers	Median Charge-to-Pay Ratio	Maximum Ratio Observed	Records Flagged as Outlier

01 Key Findings

1.1 Scale of Billing Anomalies

The median charge-to-payment ratio across all provider-DRG records is **5.52x**, meaning the typical hospital submits charges to Medicare that are 5.52 times the amount Medicare ultimately pays. While a markup above 1.0x is expected under the IPPS system, the distribution is heavily right-skewed — the maximum observed ratio of **141.52x** indicates extreme outliers that warrant immediate review. A total of **7,840 records (5.36%)** were flagged as statistical outliers using the IQR method with a domain-logic floor at 1.0x.

1.2 Financial Exposure

Across all 7,840 statistically flagged outlier records, the cumulative gap between submitted charges and Medicare payments represents a significant concentration of billing risk:

Total submitted charges (outlier records): $44.8 billion

Total Medicare payments (outlier records): $2.78 billion

Total charge-to-payment gap (outlier records): $42.1 billion

Outlier providers submit charges at an average of 16.1x Medicare reimbursement — nearly 3x the national median ratio of 5.52x.

For context, total submitted charges across all 146,251 records amount to $431.9 billion against $73.1 billion in Medicare payments — an overall ratio of 5.9x. Outlier records represent 10.4% of total charges despite comprising only 5.36% of records, confirming their disproportionate financial footprint.

1.3 Geographic Concentration

Billing anomalies are not uniformly distributed across the country. Three states account for a disproportionate share of high-ratio providers, suggesting systemic billing practices rather than isolated individual errors:

State	Finding

Nevada (NV)	Highest average charge-to-payment ratio nationally. Consistent outlier concentration across multiple DRG categories.
Florida (FL)	Largest total charge-to-payment gap of any state at $9.07 billion across outlier records — the single highest financial exposure state in the dataset.
Colorado (CO)	Third-highest average ratio. Concentrated in specific surgical DRG categories including musculoskeletal and circulatory procedures.

1.4 Procedure Category Patterns

Analysis of charge-to-payment ratios by DRG clinical category reveals that anomalies are not evenly distributed across procedure types. The three highest-ratio categories by median ratio are:

DRG Category	Finding
ENT	Highest median ratio of any DRG category. Ear, nose and throat procedures show systematic charge inflation relative to Medicare reimbursement rates.
Eye & Ear	Second-highest median ratio. Overlaps partially with ENT in procedure scope; independently confirms the pattern in head and neck billing.
Alcohol & Drug	Third-highest median ratio. Substance use disorder treatment billing shows disproportionate charge-to-payment gaps across multiple states.

02 Audit Risk Scoring Model

A composite audit risk score (0–100) was developed to rank provider-DRG records by priority for compliance review. The model is rule-based and fully explainable — each component directly maps to an observable data signal.

Component	Weight	Signal	Direction
Charge-to-payment ratio	**50 pts**	Primary billing anomaly signal	Higher ratio → higher score
Discharge volume	**35 pts**	Low volume amplifies audit risk	Lower volume → higher score
IQR outlier flag	**15 pts**	Statistical confirmation signal	Flagged = 15 pts, else 0

Risk Tier Distribution

Tier	Score Range	Records	% of Total	Distinct Providers
Critical	78–100	**7,328**	**5.0%**	**550**
High	50–78	36,056	24.7%	—
Medium	43–50	49,284	33.7%	—
Low	0–43	53,583	36.6%	—

Note: Score bin edges were derived from actual score percentiles (25th/50th/95th) rather than arbitrary round numbers, ensuring meaningful differentiation across tiers. Distinct provider count is reported for Critical tier only as this is the primary action group.

03 Recommendations

	Action
Recommendation 1	**Prioritise immediate audit review of the 550 Critical-tier providers.** These 550 distinct hospitals account for 7,328 provider-DRG records scoring in the 78–100 range. Florida represents the highest financial exposure at $9.07 billion in charge-to-payment gap and should be the first state-level focus. ENT and Eye & Ear procedure categories should be the first DRG-level focus.
Recommendation 2	**Commission a targeted procedure-level review of ENT, Eye & Ear, and Alcohol & Drug DRG categories.** These three categories show the highest median charge-to-payment ratios nationally and appear consistently across high-ratio states. The pattern suggests a possible coding or tariff issue rather than isolated provider behaviour.
Recommendation 3	**Implement a quarterly refresh of this analysis as a standing compliance dashboard.** The current model can be re-run on each annual CMS IPPS data release with no code changes. Tracking ratio trends year-over-year per provider would strengthen the model by flagging providers whose ratios are increasing, not just those currently high.
Recommendation 4	**Validate scoring weights against historical audit outcomes.** The current 50/35/15 weighting is analytically reasoned but not empirically calibrated. If CMS audit finding records are available, a logistic regression on confirmed audit cases would produce evidence-based weights and strengthen the model's defensibility in a regulatory context.

04 Methodology Note

Dataset	CMS Medicare Inpatient Hospitals by Provider and Service (IPPS), DY2023. Source: data.cms.gov. 146,251 provider-DRG records, 2,937 distinct providers across all US states and territories.
Cleaning	5-stage Python/pandas pipeline: currency column coercion ($-strip), identifier zero-padding (CCN, ZIP), text standardisation (name abbreviations, case), duplicate detection (natural key: Provider CCN x DRG), outlier separation (logical and statistical).
Ratio	Charge-to-payment ratio = Avg_Submtd_Cvrd_Chrg / Avg_Mdcr_Pymt_Amt. Rows where ratio < 1.0 (payment exceeds charge — logically impossible under IPPS) were separated as data errors and excluded from analysis.
Outlier flag	IQR method with multiplier 1.5. Lower bound floored at 1.0 by domain logic. Upper bound = Q3 + 1.5 x IQR. 7,840 records flagged (5.36%).
Risk score	Additive weighted score: ratio_score (0-50) + volume_score (0-35, inverted log-normalised) + outlier_score (0-15, binary). Maximum possible: 100. Tier bins derived from score percentiles.
Tools	Python 3 (pandas, numpy, matplotlib, seaborn), DuckDB (SQL analysis), Power BI Desktop June 2026 (dashboard). Full pipeline available at: github.com/[your-username]/cms-medicare-audit-analysis